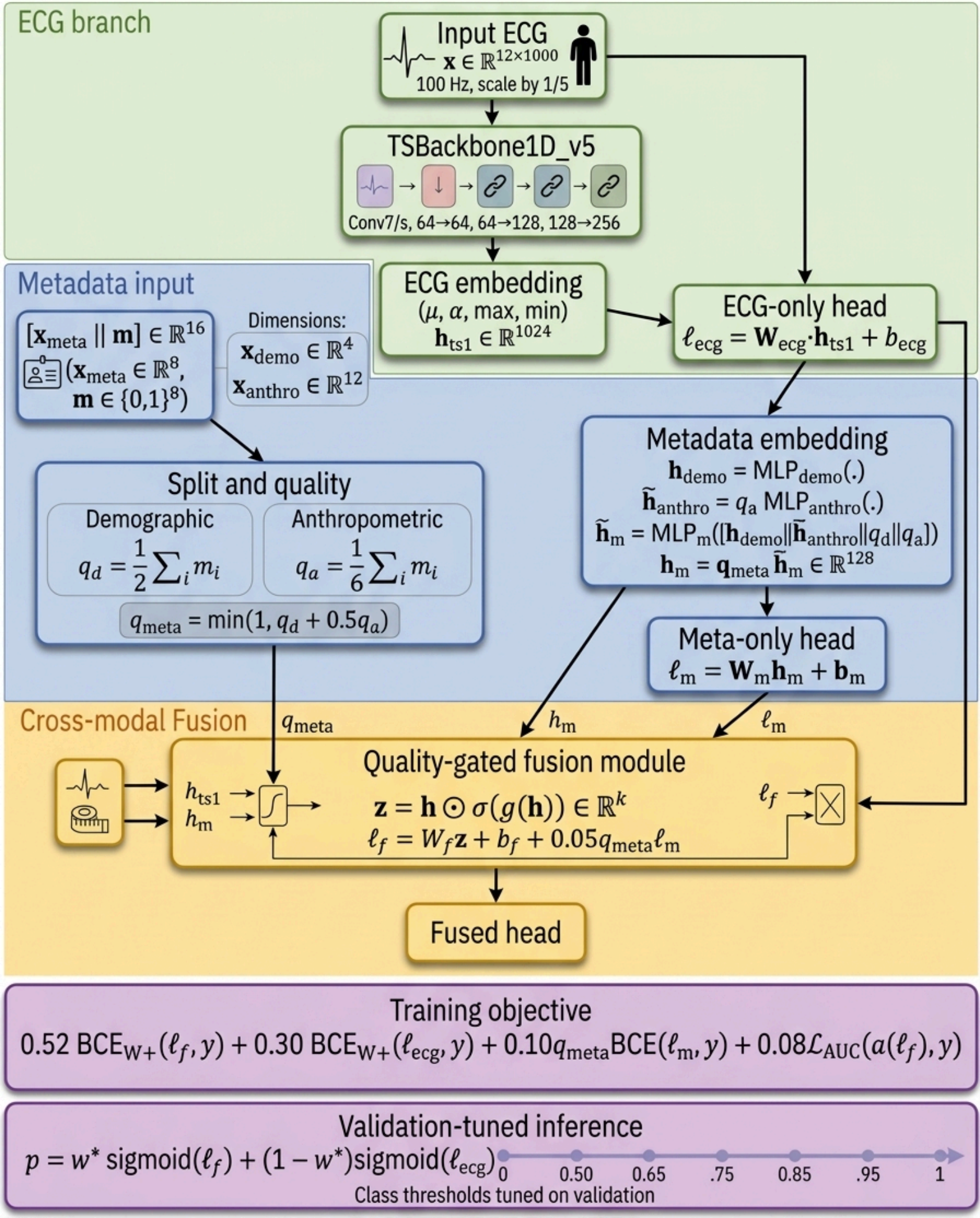
ECG branch
Input ECG
$\mathbf{x} \in \mathbb{R}^{12\times1000}$
100 Hz, scale by 1/5
TSBackbone1D_v5
Conv7/s, 64→64, 64→128, 128→256
ECG embedding
(μ, α, max, min)
$\mathbf{h}_{ts1} \in \mathbb{R}^{1024}$
ECG-only head
$\ell_{ecg} = \mathbf{W}_{ecg} \cdot \mathbf{h}_{ts1} + b_{ecg}$
Metadata input
$[\mathbf{x}_{meta} \| \mathbf{m}] \in \mathbb{R}^{16}$
($\mathbf{x}_{meta} \in \mathbb{R}^{8}$, $\mathbf{m} \in \{0,1\}^{8}$)
Dimensions:
$\mathbf{x}_{demo} \in \mathbb{R}^{4}$
$\mathbf{x}_{anthro} \in \mathbb{R}^{12}$
Split and quality
Demographic
$q_d = \frac{1}{2}\sum_i m_i$
Anthropometric
$q_a = \frac{1}{6}\sum_i m_i$
$q_{meta} = \min(1, q_d + 0.5q_a)$
Metadata embedding
$\mathbf{h}_{demo} = \mathrm{MLP}_{demo}(.)$
$\tilde{\mathbf{h}}_{anthro} = q_a\, \mathrm{MLP}_{anthro}(.)$
$\tilde{\mathbf{h}}_{m} = \mathrm{MLP}_{m}([\mathbf{h}_{demo} \| \tilde{\mathbf{h}}_{anthro} \| q_d \| q_a])$
$\mathbf{h}_{m} = \mathbf{q}_{meta}\, \tilde{\mathbf{h}}_{m} \in \mathbb{R}^{128}$
Meta-only head
$\ell_m = \mathbf{W}_m \mathbf{h}_m + \mathbf{b}_m$
Cross-modal Fusion
q_{meta}
h_m
ℓ_m
h_{ts1}
h_m
Quality-gated fusion module
$\mathbf{z} = \mathbf{h} \odot \sigma(g(\mathbf{h})) \in \mathbb{R}^{k}$
$\ell_f = W_f \mathbf{z} + b_f + 0.05 q_{meta} \ell_m$
ℓ_f
Fused head
Training objective
$0.52\,\mathrm{BCE}_{W+}(\ell_f, y) + 0.30\,\mathrm{BCE}_{W+}(\ell_{ecg}, y) + 0.10 q_{meta}\mathrm{BCE}(\ell_m, y) + 0.08\mathcal{L}_{AUC}(a(\ell_f), y)$
Validation-tuned inference
$p = w^* \,\mathrm{sigmoid}(\ell_f) + (1 - w^*)\mathrm{sigmoid}(\ell_{ecg})$
0
0.50
0.65
.75
0.85
.95
1
Class thresholds tuned on validation